



Method Validation Report

The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Rat Serum

QPS Project Number: 344-1402

Client Project Number: NA

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METHOD VALIDATION REPORT

REPORT TITLE:

The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Rat Serum

QPS PROJECT NUMBER:

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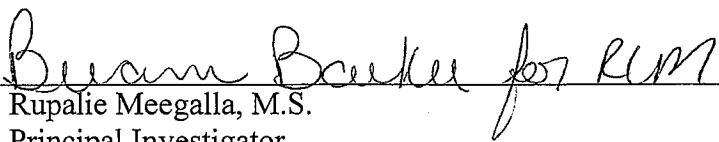
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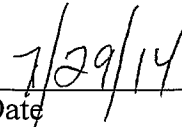
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1. SIGNATURE PAGE

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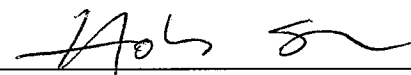


Rupalie Meegalla, M.S.
Principal Investigator
QPS, LLC

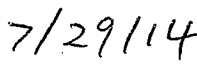


Date

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Date

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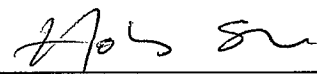
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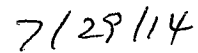
2. COMPLIANCE STATEMENT

This portion of the study was conducted in compliance with the Food and Drug Administration (FDA) Good Laboratory Practice Regulations (GLP) as set forth in Title 21 of the U.S. Code of Federal Regulations Part 58 and the FDA Guidance for Industry: Bioanalytical Method Validation, May 2001.

To the best of my knowledge this report accurately reflects the raw data. I have reviewed the study and agree that the data supports the conclusions stated herein.

 for Rupalie Meegalla

Rupalie Meegalla, M.S.
Principal Investigator
QPS, LLC



Date

3. QUALITY ASSURANCE STATEMENT

QPS: 344-1402

Protocol Title: The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Rat Serum

This QPS study has been audited and found to accurately reflect the requirements set forth by the validation plan 344-1402 and standard operating procedures and the results reported accurately reflect the source data. The following table summarizes the inspections and audits performed throughout the course of the study and the dates when each audit report and any findings were reported by QPS Quality Assurance.

Audit Description	QPS Auditor(s)	Audit Dates	Date Audit Findings Reported to QPS Principal Investigator and Management
Draft Protocol	Y. Li	27 June 2014	27 June 2014
Draft Protocol Amendment 1	Y. Li	17 July 2014	17 July 2014
Study Records	J. McClary	22 July 2014 23 July 2014	23 July 2014
Draft Report	J. McClary	24 July 2014	24 July 2014

The undersigned has assured that the QPS final report accurately reflects the source data.


Jane McClary, B.A.
QPS Quality Assurance


Date

4. VALIDATION SUMMARY FOR ANTI-LT3114 ANTIBODIES

A quasi-quantitative immunogenicity assay to detect the presence of anti-LT3114 antibodies in rat serum samples was validated. Experiments were conducted at QPS to assess assay precision, screening and confirmation assay cut-point factor for rat serum, matrix selectivity, drug tolerance, and PC stability. The results are provided in the following summary table.

Report Title	The Validation of an ECL Assay for the Detection of Anti-Drug Antibodies (ADA) Directed Against LT3114 in Rat Serum		
Report Number	QPS 344-1402		
Analyte Name	Anti-LT3114 antibodies in rat Serum		
Negative Control (NC)	A pool of rat serum prepared from at least 20 individual lots		
Positive Controls (PC)	Whole LT3114 anti-idiotypic rabbit positive serum		
Minimum Required Dilution (MRD) and Sample Volume:	MRD = 1:90 (after mixing with master mix) Minimal 10 µL for screening Minimal 20 µL for confirmation		
Analytical Method Type	ECL		
Detection Instrument	MSD Sector Imager 6000		
Analytical System Software and Version	Discovery Workbench™ 2006 MSD 3.0.17_3		
Intra-Run Precision (%CV) of Whole LT3114 anti-idiotypic rabbit positive serum (HPC 1:100, MPC 1:400 & LPC 1:3200 dilutions in rat serum pool)	HPC	MPC	LPC
	7.7%	7.8%	5.6%
Inter-Run Precision (%CV) of Whole LT3114 anti-idiotypic rabbit positive serum (HPC 1:100, MPC 1:400 & LPC 1:3200 dilutions in rat serum pool)	HPC	MPC	LPC
	5.8%	7.3%	5.6%
Screening Cut Point Factor (N)	8.83		
Calculation of Screening Cut Point	Plate Mean NC RU + 8.83		
Confirmatory Cut Point Factor	36.7%		
Matrix Selectivity (Recovery)	HPC(1:100 diluted positive serum): 100% lots tested have recovery between 75% and 125% LPC(1:3200 diluted positive serum): 100% lots tested have recovery between 75% and 125%		

Drug Tolerance	HPC can tolerate at least 200µg/mL free LT3114 drug and LPC can tolerate at least 10µg/mL free LT3114 drug in rat serum.
Room Temperature Stability of PC in rat serum (HPC, MPC & LPC)	24 hours at room temperature
Freeze/Thaw Stability of PC in rat serum	5 Cycles at -70°C

5. OBJECTIVE

The purpose of this study was to validate an ECL method to detect the presence of anti-LT3114 in rat serum samples.

6. ABBREVIATIONS AND DEFINITIONS

This section provides abbreviations and definitions of terms and concepts that may be commonly used throughout this report.

ADA	Anti-Drug Antibody
PC	Positive Control
HPC	High Positive Control
MPC	Mid Positive Control
LPC	Low Positive Control
MRD	Minimum Required Dilution
RTSE	Rat Serum
MSD	Meso Scale Discovery
NA	Not Applicable
NC	Negative Control
NR	Not Reported
RU	Response Unit
SD	Standard Deviation
SOP	Standard Operating Procedure

Precision	$\% \text{Coefficient of Variation (\%CV)} = 100 \times \frac{\text{Standard Deviation}}{\text{Mean Value}}$
------------------	--

One pooled rat serum sample was used as negative control (NC) for each validation run. Whole LT3114 anti-idiotypic rabbit positive serum was diluted in to rat serum at 1:100 for HPC, 1:400 for MPC, and 1:3200 for LPC and used as positive controls (PC) in this qualitative assay.

All statistics in the data tables are calculated by Watson IRM 7.4.1 or based on the “precision as displayed” option of Microsoft® Excel (e.g., Mean, SD, %CV, %Diff).

7. ACCEPTANCE CRITERIA

The validation acceptance criteria and the statistical data were determined according to QPS validation plan 344-1402 and QPS-TLM-001. The acceptance criteria listed below is taken from the validation study plan.

Each control sample will be diluted in the same manner analogous to how unknowns will be treated. Samples and PCs are added to duplicate wells.

For NCs, PCs and samples, the %CV from duplicate measurements should not be greater than 25%. Samples with a duplicate %CV greater than 25% will be rejected and will not be included in statistical analyses.

At least two-thirds of NCs should be reportable.

The mean values of the ECL responses of NC and three levels of PCs must be in the order of: NC < LPC < MPC < HPC.

For the plate to be accepted at least 50% of the PCs at each level should be reportable and at least two-thirds of all PCs should be reportable.

Method acceptance criteria:

Intra-assay precision: The intra-assay precision will be determined from the NC and the PCs on the assay plate. The mean and %CV of the response unit (RU) from the PCs and NC will be evaluated. The %CV of the NC and PCs at each level must be $\leq 25\%$.

Inter-assay precision: The inter-assay precision will be determined from all acceptable validation runs except intra-assay precision runs. The mean and %CV of the response unit (RU) from the PCs and NC will be evaluated. The %CV of the NC and PCs at each level must be $\leq 25\%$.

Matrix Specificity (Recovery): The % recovery of the mean response of PC-spiked individual lots will be assessed. Two-thirds of the tested lots should be within the acceptable range (75-125% recovery).

Drug tolerance test: Observations will be made as to which concentration of drug inhibits the ability of the assay to detect the PC at each level.

Freeze/thaw stability: The samples are considered stable if the mean RU value of 5 freeze/thaw cycle samples are within $\pm 25\%$ difference of the mean RU value of PCs that were thawed once immediately before analysis.

Room temperature stability: The samples are considered stable if the mean RU value of the samples maintained at room temperature for 24 hours and 6-8 hours are within $\pm 25\%$ difference of the mean RU value of untreated PCs.

8. MATERIALS

8.1. Sample-Handling Equipment

Equivalent equipment may be substituted on an as-needed basis

Plate Sealer: Costar Cat.# 3095

25-100 mL Reagent Reservoir: Costar Cat. # 4870

96-well format micro titer/dilution tubes (VWR Cat. # 46300-536)

Polypropylene dilution plate (Nunc Cat. # 267385)

0.5-50 mL polypropylene tubes, VWR Scientific

500 or 1000 mL wash bottle

Graduated Cylinder

Multi-tube vortexer: VWR Scientific Products

Plate Shaker

Microplate reader: Meso Scale Discovery Sector Imager 6000

Balance

8.2. Analytical Equipment

Microplate reader: Meso Scale Discovery Sector Imager 6000

9. METHODS

9.1. Key Reagent Information

Pooled normal rat serum was used as the negative control. Whole LT3114 anti-idiotypic rabbit positive serum (Lot# R-2265-B2-11/13, Exp. 5/21/16), 2 vials were received on ice from Lpath, one vial was stored at 4°C per client's communication and used for method development. Other vial was aliquoted and stored at -20°C for regulated studies as long term storage.

9.2. Critical Reagents

Drug Name: LT3114(HU-anti-LPA antibody)
Source: Lpath
Concentration: 19.33 mg/mL
Lot Number: LP121416
Expiration Date: 30 Apr 2016 (Retest date not provided, assigned per QPS SOP)
Storage Conditions: -70°C
See CoA in [Appendix A](#)

9.3. Positive Control Source Information

Name: Whole LT3114 anti-idiotypic rabbit positive serum
Source: Lpath
Concentration: NA
Lot Number: R-2265-B2-11/13
Expiration Date: 21-Jul-2014 at 4°C
Storage Conditions: -20°C
See [Appendix B](#)

9.4. Analytical Method Summary

Rat serum samples, positive controls and negative controls were diluted 1:30 in assay buffer and then incubated at room temperature for 60±10 minutes in the presence of Biotinylated LT3114 and Ruthenylated-LT3114 with additional dilution of 1:3. The antibodies (IgG, IgM, etc.) resulted in the bridging of biotinylated and ruthenylated-LT3114. After incubation, 50 µL of each mixture is added to the wells of the Streptavidin coated MSD plate and allowed to bind for 60±10 minutes. In the presence of tripropylamine-containing read buffer, ruthenium produces a chemiluminescent signal that is triggered when voltage is applied. The resulting chemiluminescence is measured in response unit (RU) on SECTOR Imager 6000 instrument.

9.5. Data Acquisition

Equipment: MSD Sector Imager 6000
Program: Discovery Workbench™ 2006 MSD 3.0.17_3

10. RESULTS AND DISCUSSION

[Table 1](#) lists all the validation runs performed.

10.1. Precision

Precision of the method was expressed as the coefficient of variation (%CV) from analyzing replicates of HPC, MPC and LPC. The mean RU of PCs and the mean RU of NC in each run was evaluated. The assay will be considered precise if the %CV of HPC, MPC and LPC from each set of PC (intra-run) and all accepted runs (inter-run) are less than 25.0%.

The intra-run precision was determined from N=5, duplicate readings and N≥5, duplicate readings of HPC (1:100), MPC (1:400) and LPC (1:3200) in rat serum on intra-run plate. The precision of intra run for HPC, MPC, LPC and NC are 7.7%, 7.8%, 5.6%, and 5.0% respectively ([Table 2](#)).

The inter-run precision of the assay was determined from plate controls, PC (N≥2, duplicate readings) and NC (N=3, duplicate readings) on each plate. The mean RU of NC and PCs from all accepted validation runs (excluding intra-assay precision runs) were included and a minimum of 6 validation runs over at least 3 days were evaluated.

The results for inter-assay precision for HPC, MPC, LPC and NC are 5.8%, 7.3%, 5.6%, and 7.2% respectively ([Table 3](#)). Both intra-run and inter-run precisions met the acceptance criteria.

10.2. Screening Cut Point and Confirmatory Cut Point Determination

The variability of antibody response for 30 rat serum samples not exposed to LT3114 was evaluated to determine the assay cut point. The rat serum samples were diluted at 1:90 (MRD) before being assayed. The non-parametric cut point was determined from the instrument signal of the negative control and individual matrix responses, based on 95% upper confidence limit according to SOP, QPS-TLM-001. Data showed normal distribution on all three days of analysis. All outlier values were reported, however, values were excluded from statistical calculations. The details of the statistical evaluation were listed in [Appendix C](#).

A cut point factor was calculated by deducting the negative control mean from plate cut point by the mean response of negative control of each plate excluding the statistical outlier, if any. Screening data from each of the three days were found normally distributed using JMP statistical software. The average calculated plate cut point factor for the screening runs of the rat serum is 8.83 ([Table 4a](#)).

The confirmatory cut point was validated to determine whether a positive response for ADA was specific to LT3114. Thirty individual lots of rat serum, as well as high and low PCs (HPC 1:100 and LPC 1:3200) were tested in the presence of 100µg/mL LT3114 as well as no drug. For each sample, the percent of signal inhibition for each of the individual matrix samples was calculated. Box-Plot approach was used to define outliers, described as $Q3 + 1.5 * (Q3 - Q1)$ and $Q1 - 1.5 * (Q3 - Q1)$, where Q3, Q1, and (Q3-Q1) are the 75th percentile, 25th percentile and interquartile range, respectively. High and low outliers were identified by JMP statistical software.

After outliers were excluded, the mean and SD of the inhibition ratios was also determined and used to calculate confirmatory cut point as following:

Confirmatory cut point = Mean % signal inhibition + 3.09*SD

Calculated confirmatory cut point is 36.7% and this confirmatory assay cut point was accepted if the LT3114 spiked high and low PC (1:100 and 1:3200, respectively) samples run as controls had a % inhibition greater than the calculated confirmatory cut point. In the study, the % inhibition is 99.4% for HPC and 81.8% for LPC, which are both higher than the calculated confirmatory cut point 36.7%. The results for the confirmatory cut points for rat serum and PC performance can be found in (Table 4b) and (Table 5).

10.3. Matrix Selectivity (Recovery)

Matrix components can inhibit the ADA from binding to the capture and/or detection reagents under assay conditions. To evaluate matrix interference, 12 individual rat serum lots, as well as the pooled rat serum (NC) were spiked with HPC and LPC ($n \geq 1$) and analyzed at MRD 1:90 in the assay. Percent recovery will be calculated for each individual rat serum lot as:

$\% \text{Recovery} = (\text{Mean Response of Spiked Lot}) / (\text{Mean Response of Spiked NC}) \times 100\%$

100% of the lots tested recovered within 75-125% when spiked with both levels of PCs, HPC (1:100) and LPC (1:3200). (Table 6)

10.4. LT3114 Drug Tolerance

In the Drug Tolerance test, the concentration of LT3114 that inhibits the ability to detect positive controls was assessed. High and low control PCs (HPC 1:100 / LPC 1:3200) and NC were incubated in the absence and presence of LT3114 at 200, 100, 50, 25, 10, 5, 0 µg/mL for ≥ 30 minutes prior to be evaluated in the assay.

The highest concentration of LT3114 that still appears positive or greater than the plate cut point for each positive control was identified. Results indicate that HPC (1:100 positive sera) can tolerate at least 200 µg/mL LT3114 and LPC(1:3200) can tolerate at least 10 µg/mL LT3114 (Table 7).

10.5. Stability

10.5.1. Freeze/Thaw Stability of Positive Control

Freeze/thaw stability of LT3114 anti-idiotypic rabbit positive serum in rat serum was used as the surrogate to evaluate anti-LT3114 stability in the matrix after 5 freeze/thaw cycles. HPC(1:100), MPC (1:400), and LPC (1:3200) ($n=3$) were frozen for at least 24 hours at -70°C and then subjected to 5 freeze/thaw cycles. Samples were allowed to thaw at room temperature under normal assay conditions for at least one hour and then frozen back to -70°C or colder for at least 12 hours. The treated PC as well as untreated PC (1 freeze/thaw cycle) was used to analyze in the assay to determine the FT stability. The results showed that all 3 levels of PC met the acceptance criteria with % Diff at -6.6%, -6.9% and -13.2% for

HPC, MPC and LPC respectively (Table 8), suggesting that the positive control is stable for at least 5 freeze/thaw cycles at -70°C.

10.5.2. Room Temperature Stability of Positive Control

Room temperature stability of LT3114 anti-idiotypic rabbit positive serum in rat serum was used as the surrogate to evaluate anti LT3114 stability in the matrix for the time that the samples were at room temperature during sample handling and processing. HPC(1:100), MPC (1:400), and LPC (1:3200) (n=3) were thawed at room temperature under normal assay conditions and stored at room temperature for at least 7 and 24 hours. After incubation at room temperature, the PCs were frozen back at 70°C for at least 12 hours and then tested with the untreated PC (not exposed to room temperature as the treated PC).

Initially, 24 hour room temperature stability was evaluated. HPC did not meet the acceptance criteria with %Diff at -36.6% when compared to untreated PC samples, but the other 2 PC levels were within $\pm 25\%$ when compared to untreated samples (Table 9A). The test was repeated for HPC and LPC at N=6. The results showed that HPC and LPC met the acceptance criteria with % Diff at 18.5% and -22.9% respectively (Table 9B).

Seven (7) hour room temperature stability was also evaluated at 3 PC levels. The results showed that all 3 levels of PC met the acceptance criteria with % Diff at 7.1%, -8.5% and -17.1% for HPC, MPC and LPC respectively (Table 9C).

10.6. Sample Analysis Acceptance Criteria Determination

Based on the results from the validation, the following are the sample analysis acceptance criteria.

For a sample analysis run, PCs (N=2, duplicate readings) and NC (N=3, duplicate readings) will be included on each plate as controls.

Run acceptance criteria:

1. For NC and PCs, the %CV from duplicate measurements should not be greater than 25% unless both measurements are below the plate cut point. Otherwise, the data is not reportable.
2. At least two-thirds of NCs should be reportable.
3. Plate NC Mean RU for each run must be $\leq$ (Inter-assay Mean NC x 1.5).
4. The mean values of the ECL responses of NC and three levels of PCs must be in the order of: NC < LPC < MPC < HPC.
5. For the plate to be accepted at least 50% of the PCs at each level should be reportable and at least two-thirds of all PCs should be reportable.

Study samples at MRD (1:90) from each sample will be tested first by a screening assay. The %CV of RU from duplicate measurements should be $\leq 25\%$ if at least one of the responses is $\geq$ the cut point on the same plate. Otherwise, the data is not reportable and the sample will be re-assayed.

- (1) If the mean RU of a sample is < plate cut point, the sample is reported negative.
- (2) If the mean RU of a sample yields a mean RU $\geq$ plate cut point, the sample should evaluate in the confirmation assay
- (3) All screened positive samples will be analyzed both spiked and un-spiked with 100 μ g/mL LT3114.
- (4) For the confirmatory results to be accepted, at least 50% of the confirmatory PCs (HPC 1:100 and LPC 1:3200) at each level should be acceptable (i.e., reportable, with $\geq 36.7\%$ %inhibition).
- (5) Samples with %inhibition less than the validated CCP (36.7%) will be reported as “Negative” for the confirmatory result.
- (6) Samples with an unspiked mean response equal to or greater than the plate cut point and %inhibition greater than or equal to the validated CCP (36.7%) will be reported as “Positive” for the confirmatory result.
- (7) If the unspiked mean response is below the plate cut point with %inhibition $\geq 36.7\%$, the sample will be reassayed using the confirmation assay and the results of the reassay will be reported.

11. STUDY SPECIFIC DEVIATIONS OR OBSERVATIONS

DEV-14-357:

MPC was not tested as confirmation PC during confirmatory cut point determination according to 344-1402 validation protocol. Both HPC (1:100) and LPC (1:3200) were tested and they all showed the percent inhibition greater than the calculated confirmatory cut point which is 36.7%. HPC used in these runs (Run 5 & 6) gave mean inhibition of 99.4% for the HPC, and LPC used in these runs (Run 5 & 6) gave mean inhibition of 81.8%, showed that these data are sufficient to accept the confirmatory assay cut point and met the acceptance criteria defined in the validation protocol. This deviation does not impact the validation test results for confirmatory cut point determination.

12. CONCLUSION

An ECL method (QPS method 344-1402) has been validated for the detection of anti-LT3114 in rat serum.

13. REFERENCES

QPS Validation Plan

QPS Validation Plan 344-1402

SOPs

QPS GN-014-12A: Archives

QPS DM-010-01 Immune Response Module Using Watson LIMS

QPS-TLM-001-00 Validation of Immunogenicity Assays

Study Notebooks

Validation Notebook QPS 344-1402

Client 344 Compound Receipt Notebook

Client 344 Assay Notes

Client 344 Reagent Receipt Notebook

White Paper

Shankar G., Devanarayan V., Amaravadi L., et al. Recommendations for the Validation of Immunoassays Used for Detection of Host Antibodies Against Biotechnology Products. J. Pharm. Biomed. Anal. 2008, 48, 1267-1281.

14. ARCHIVE STATEMENT

All raw data (including electronic SoftmaxPro files), a copy of this bioanalytical report and required supporting information for this study will be held under the control of QPS, LLC, 3 Innovation Way, Suite 240, Newark DE, 19711, USA. Study records will be archived according to the most current version of SOP GN-014, "Archives".

15. TABLES

Table 1 Validation Run Log

Run Number	Run ID	Validation Items	Rat Serum Lots Tested	Run Status
1	140709-3441402-RTSE-MM-1A-001	Screening (Day 1)	1-15 (Male) & 16-30 (Female)	Accepted
2	140709-3441402-RTSE-MM-2B-002	Intra-Day / 24hr RT Stability	NA	Accepted ^a
3	140710-3441402-RTSE-MM-1A-003	Matrix Selectivity	1-6 (Male) & 16-21 (Female)	Accepted
4	140710-3441402-RTSE-RLM-1A-004	Screening (Day 2)	1-15 (Male) & 16-30 (Female)	Accepted
5	140710-3441402-RTSE-RLM-2B-005	LT314 Confirmation Plate 1	1-8 (Male) & 17-23 (Female)	Accepted
6	140710-3441402-RTSE-RLM-3C-006	LT314 Confirmation Plate 2	9-16 (Male) & 24-30 (Female)	Accepted
7	140711-3441402-RTSE-RLM-1A-007	Screening (Day 3)	1-15 (Male) & 16-30 (Female)	Accepted
8	140711-3441402-RTSE-RLM-2B-008	Drug Tolerance / FT Stability	NA	Accepted
9	140711-3441402-RTSE-RLM-3C-009	Drug Tolerance / 24hr RT Stability	NA	Accepted ^b
10	140716-3441402-RTSE-RLM-1A-010	6-8 hr RTS / 24 hr RTS repeated N=2 for HPC & LPC	NA	Accepted

Run Status: Run accepted / rejected according to QPS VA protocol 344-1402

a: Room Temperature test rejected due to analytical error

b: 24 hour room temperature stability failed at HPC level

Table 2 Intra-Run Precision of anti-LT3114 anti-idiotypic rabbit antibody Positive Control in Rat Serum

Number (N=)	HPC(1:100)	CV%	MPC(1:400)	CV%	LPC(1:3200)	CV%	NC	CV%
1	10636	3.1	2347	0.2	293.5	0.2	70.50	3.0
2	8565	2.8	1913	0.7	290.0	1.5	70.00	2.0
3	10176	1.8	2010	0.7	290.5	0.7	79.00	9.0
4	10480	0.5	2077	0.9	320.0	2.2	75.50	2.8
5	10598	1.4	2080	1.8	272.5	2.3	74.50	0.9
6	10024	0.9	1914	2.4	308.5	1.1	NA	NA
N	6		6		6		5	
Mean	10080	1.8	2057	1.1	295.8	1.3	73.9	3.5
SD	780		160		16.5		3.73	
%CV	7.7		7.8		5.6		5.0	

Run ID: 140709-3441402-RTSE-MM-2B-002

Table 3 Inter-Run Precision of anti-LT3114 anti-idiotypic rabbit antibody Positive Control in Rat Serum

Run Date	Run	HPC Mean	MPC Mean	LPC Mean	Mean NC
9-Jul-2014	1	9854	2270	304.5	85.00
		9856	1763	303.0	86.50
					77.50
10-Jul-2014	3	10040	1996	307.0	84.50
		10123	1994	298.0	86.50
					83.50
10-Jul-2014	4	9552	1964	294.5	80.50
		9393	1843	273.5	80.50
					71.50
10-Jul-2014	5	8757	1788	280.0	76.50
		8817	1754	259.0	78.00
					63.50
10-Jul-2014	6	8754	1793	272.0	72.50
		9091	1780	264.0	73.00
					66.00
11-Jul-2014	7	9442	1994	301.5	80.00
		9240	1776	264.5	80.50
					77.50
11-Jul-2014	8	8901	1851	283.5	76.50
		8486	1749	267.5	80.00
					72.00
11-Jul-2014	9	8401	1784	278.5	80.00
		8590	1755	264.0	77.00
					77.00
16-Jul-2014	10	9358	1942	290.0	77.50
		9426	1899	284.5	75.00
					73.00
N		18	18	18	27
Mean		9227	1872	282.8	77.5
SD		535	136	15.9	5.61
%CV		5.8	7.3	5.6	7.2

Table 4a Screening Assay Cut Point Evaluation of Individual Rat Serum Samples in Anti-LT3114 Detection Assay

Lot ID	Lot Number	Gender	Day 1-Run 1		Day 2-Run 4		Day 3-Run 7	
			RU	CV%	RU	CV%	RU	CV%
Lot 01	RAT237202	Male	93.50	6.8	74.50	6.6	81.50	6.1
Lot 02	RAT237203	Male	87.50	8.9	80.00	1.8	79.50	6.2
Lot 03	RAT237204	Male	83.00	1.7	77.50	0.9	78.50	0.9
Lot 04	RAT237205	Male	81.00	1.7	76.50	0.9	80.50	0.9
Lot 05	RAT237206	Male	84.50	5.9	75.50	6.6	83.00	0.0
Lot 06	RAT237207	Male	84.50	2.5	79.00	0.0	80.00	1.8
Lot 07	RAT237208	Male	84.00	3.4	76.00	0.0	79.50	4.4
Lot 08	RAT237209	Male	86.50	0.8	74.50	2.8	80.50	0.9
Lot 09	RAT237210	Male	87.50	4.0	79.00	0.0	80.00	3.5
Lot 10	RAT237211	Male	90.00	0.0	84.50	0.8	82.50	2.6
Lot 11	RAT237212	Male	98.00	0.0	85.00	5.0	92.00 ^a	3.1
Lot 12	RAT237213	Male	90.50	0.8	83.50	0.8	80.50	2.6
Lot 13	RAT237214	Male	87.50	5.7	82.50	4.3	84.00	1.7
Lot 14	RAT237215	Male	85.00	6.7	76.50	2.8	81.50	0.9
Lot 15	RAT237216	Male	80.50	4.4	72.50	6.8	78.50	0.9
Lot 16	RAT237220	Female	93.50	0.8	80.00	0.0	87.50	0.8
Lot 17	RAT237221	Female	90.00	3.1	77.00	3.7	78.50	0.9
Lot 18	RAT237222	Female	86.50	2.5	81.50	2.6	80.00	0.0
Lot 19	RAT237223	Female	87.50	2.4	78.00	1.8	82.00	1.7
Lot 20	RAT237224	Female	97.00	1.5	85.50	0.8	88.00	0.0
Lot 21	RAT237225	Female	82.50	6.0	70.00	0.0	79.00	1.8
Lot 22	RAT237226	Female	80.50	2.6	73.50	1.0	76.50	0.9
Lot 23	RAT237227	Female	77.50	0.9	73.50	2.9	76.00	1.9
Lot 24	RAT237228	Female	81.50	0.9	76.00	1.9	79.00	3.6
Lot 25	RAT237229	Female	85.50	0.8	78.00	1.8	82.00	0.0
Lot 26	RAT237230	Female	241.0 ^a	5.3	193.5 ^a	0.4	239.0 ^a	2.4
Lot 27	RAT237231	Female	95.50	3.7	87.00	3.3	92.50 ^a	0.8
Lot 28	RAT237232	Female	88.50	0.8	79.00	3.6	81.00	0.0
Lot 29	RAT237233	Female	77.50	0.9	72.50	2.9	74.00	5.7
Lot 30	RAT237234	Female	76.50	0.9	72.00	2.0	77.00	1.8
N			29		29		27	
Mean RU of Individual Rat Serum			86.33		77.95		80.39	
SD			5.68		4.42		3.06	
%CV			6.6		5.7		3.8	
Daily Screen Cut Point			95.67		85.22		85.42	
Negative Control Mean			83.00		77.50		79.33	
Daily screening Cut Point Factor			12.67		7.72		6.09	
Screening Cut Point Factor (N)			8.83					

Run IDs: 140709-3441402-RTSE-MM-1A-001, 140710-3441402-RTSE-RLM-1A-004 and 140711-3441402-RTSE-RLM-1A-007

a: Statistical outliers excluded from the screening cut point assessment by Box-Plot (Using JMP statistical software)

Daily Cut Point: Mean RU of individual matrix samples from one day of Analysis + (1.645 × SD of individual samples)

Normalization factor: Cut point - Negative Control Mean

Table 4b Confirmatory Assay Cut Point Evaluation of Individual Rat Serum Samples in Anti-LT3114 detection Assay

Lot ID	Lot Number	Gender	Run No.	Unspiked	%CV	Spiked ^a	%CV	% Inhibition ^b
Lot 01	RAT237202	Male	5	75.50	4.7	59.00	9.6	21.9
Lot 02	RAT237203	Male	5	76.00	0.0	58.00	4.9	23.7
Lot 03	RAT237204	Male	5	77.50	2.7	60.00	0.0	22.6
Lot 04	RAT237205	Male	5	73.50	1.0	59.50	1.2	19.1
Lot 05	RAT237206	Male	5	74.00	3.8	57.50	8.6	22.3
Lot 06	RAT237207	Male	5	74.00	5.7	52.00	2.7	29.7
Lot 07	RAT237208	Male	5	76.50	0.9	54.50	9.1	28.8
Lot 08	RAT237209	Male	5	73.50	4.8	54.50	1.3	25.9
Lot 09	RAT237210	Male	6	71.50	4.9	53.50	4.0	25.2
Lot 10	RAT237211	Male	6	75.00	0.0	55.50	1.3	26.0
Lot 11	RAT237212	Male	6	78.00	3.6	58.50	1.2	25.0
Lot 12	RAT237213	Male	6	74.00	3.8	56.50	1.3	23.7
Lot 13	RAT237214	Male	6	75.50	6.6	53.50	4.0	29.1
Lot 14	RAT237215	Male	6	71.50	1.0	51.50	4.1	28.0
Lot 15	RAT237216	Male	6	72.00	5.9	53.50	11.9	25.7
Lot 16	RAT237220	Female	6	73.50	2.9	54.00	0.0	26.5
Lot 17	RAT237221	Female	5	73.00	1.9	58.50	3.6	19.9
Lot 18	RAT237222	Female	5	78.50	0.9	58.50	3.6	25.5
Lot 19	RAT237223	Female	5	77.50	0.9	62.00	4.6	20.0
Lot 20	RAT237224	Female	5	83.50	2.5	62.00	0.0	25.8
Lot 21	RAT237225	Female	5	68.00	2.1	57.00	0.0	16.2
Lot 22	RAT237226	Female	5	69.00	0.0	59.50	5.9	13.8
Lot 23	RAT237227	Female	5	69.50	3.1	59.50	1.2	14.4
Lot 24	RAT237228	Female	6	77.50	6.4	60.00	2.4	22.6
Lot 25	RAT237229	Female	6	74.50	0.9	58.50	6.0	21.5
Lot 26	RAT237230	Female	6	175.5	2.8	60.50	5.8	65.5 ^c
Lot 27	RAT237231	Female	6	82.50	2.6	57.00	2.5	30.9
Lot 28	RAT237232	Female	6	73.50	2.9	56.50	8.8	23.1
Lot 29	RAT237233	Female	6	70.50	3.0	56.50	6.3	19.9
Lot 30	RAT237234	Female	6	68.00	4.2	55.00	12.9	19.1

N 29

Mean %Inhibition 23.3

SD 4.35

Mean % Inhibition + (3.09*SD) 36.7

Confirmatory Cut Point (%) 36.7

Confirmation Runs: 140710-3441402-RTSE-RLM-2B-005, 140710-3441402-RTSE-RLM-3C-006

- a: Samples were incubated in the presence of 100 µg/mL LT3114.
- b: $100 \times (1 - \text{RU of LT3114 Spiked Sample} / \text{RU of Unspiked Sample})$
- c: Data excluded from confirmatory cut point calculation due to lot being identified as outlier by Box-Plot Analysis (QPS Protocol 344-1402)

Table 5 Confirmation Inhibition of PC (LT3114 anti-idiotypic rabbit antibody) with and without LT3114 Spike (100 µg/mL)

Run Number	HPC (1:100)					LPC (1:3200)				
	NS	%CV	Spiked	%CV	%Inhibition	NS	%CV	Spiked	%CV	%Inhibition
5	8757	2.8	52.50	9.4	99.40	280.0	0.5	52.00	5.4	81.43
	8817	1.3	50.00	5.4	99.43	259.0	0.5	48.50	1.5	81.27
6	8754	0.4	50.00	5.7	99.43	272.0	2.6	47.00	9.0	82.72
	9091	0.8	47.50	13.4	99.48	264.0	1.6	48.50	4.4	81.63
N	4		4		4	4		4		4
Mean	8855		50.00		99.4	268.8		49.00		81.8
SD	160		2.04		0.0332	9.22		2.12		0.655
%CV	1.8		4.1		0.0	3.4		4.3		0.8

Confirmation Runs: 140710-3441402-RTSE-RLM-2B-005, 140710-3441402-RTSE-RLM-3C-006

%Inhibition = 100*(1 - LT3114 spiked RU/ NS RU)

Spiked: Samples were incubated in the presence of 100 µg/mL LT3114 before being processed.

NS: Non-spiked sample

Table 6 Matrix Selectivity (Recovery) of anti-LT3114 anti-idiotypic rabbit antibodies in Individual Rat Serum

Lot ID	Lot Number	Unspiked RU	Spiked with 1:100(HPC) RU	% Recovery	Spiked with 1:3200(LPC) RU	% Recovery
NC	RLM140620-01	83.00	14656	NA	348.5	NA
MKSE 01	RAT237202	84.50	12689	86.6	331.0	95.0
MKSE 02	RAT237203	84.50	15258	104.1	324.5	93.1
MKSE 03	RAT237204	84.50	13772	94.0	337.0	96.7
MKSE 04	RAT237205	84.50	13400	91.4	329.5	94.5
MKSE 05	RAT237206	88.00	13839	94.4	314.0	90.1
MKSE 06	RAT237207	84.50	14819	101.1	345.5	99.1
MKSE 16	RAT237220	87.00	14797	101.0	363.5	104.3
MKSE 17	RAT237221	83.00	13383	91.3	349.5	100.3
MKSE 18	RAT237222	84.00	12835	87.6	341.5	98.0
MKSE 19	RAT237223	88.00	14485	98.8	320.5	92.0
MKSE 20	RAT237224	88.50	13229	90.3	345.5	99.1
MKSE 21	RAT237225	84.00	11925	81.4	325.0	93.3

RU: Response Unit

NA: Not Applicable

Run ID: 140710-3441402-RTSE-MM-1A-003

% Recovery = (Mean Spiked Lot RU / Mean Spiked NC RU) x 100%

Spiked sample: Spiked with HPC and LPC before MRD

Table 7 LT3114 Drug Tolerance in the Anti-LT3114 Antibody Detection Assay in Rat Serum using anti-LT3114 anti-idiotypic rabbit antibody

LT3114 (µg/mL)	Run 008			Run 009		
	HPC (1:100)	LPC (1:3200)	NC	HPC (1:100)	LPC (1:3200)	NC
0	8289	232.5	80.50	7982	219.0	83.50
5	407.0	103.0	78.00	382.0	94.00	77.00
10	226.5	97.00	74.00	221.5	90.50	76.50
25	174.0	87.50	78.00	173.0	85.00	73.00
50	155.5	78.50	74.00	150.5	80.50	71.50
100	124.0	75.50	73.50	124.5	70.50	70.50
200	91.50	68.50	65.50	93.50	68.00	66.00
Mean RU of Plate NC	76.2			78.0		
Plate Cut Point	85.0			86.8		

Plate Cut Point: Mean RU of Plate NC + Normalization Factor

Run IDs: 140711-3441402-RTSE-RLM-2B-008, 140711-3441402-RTSE-RLM-3C-009

Highest level of LT3114 where PC levels are $\geq$ plate cut point are bolded

Table 8 Freeze/Thaw Stability of anti-LT3114 anti-idiotypic rabbit antibodies in Rat Serum

PC Level	Treated (FT^a 5 Cycles) / RU values			Untreated /RU Values			% Difference
HPC 01	8358	Mean	8432	8901	Mean	9024	-6.6
HPC 02	8500	SD	71.2	8486	SD	609	
HPC 03	8438	%CV	0.8	9686	%CV	6.7	
MPC 01	1743	Mean	1715	1851	Mean	1842	-6.9
MPC 02	1731	SD	38.0	1749	SD	88.4	
MPC 03	1672	%CV	2.2	1925	%CV	4.8	
LPC 01	237.0	Mean	244	283.5	Mean	281	-13.2
LPC 02	250.5	SD	6.83	267.5	SD	11.8	
LPC 03	245.5	%CV	2.8	290.5	%CV	4.2	

PC: LT3114 anti-idiotypic rabbit antibody dilutions in pooled Rat Serum (HPC 1:100, LPC 1:400 & LPC 1:3200)

Run ID: 140711-3441402-RTSE-RLM-2B-008

%Difference: $100 * ((\text{Mean 5FT treated PC} - \text{Mean untreated PC}) / (\text{Mean untreated PC}))$

RU: Response Unit

a: Freeze temperature -70°C

Table 9A Room Temperature (24 hour) Stability of anti-LT3114 anti-idiotypic rabbit antibodies in Rat Serum (Initial)

PC Level	Treated (24hr RT) / RU values			Untreated /RU Values			% Difference
HPC 01	7279	Mean	5501	8401	Mean	8671	-36.6
HPC 02	4641	SD	1540	8590	SD	318	
HPC 03	4582	%CV	28.0	9022	%CV	3.7	
MPC 01	1546	Mean	1541	1784	Mean	1798	-14.3
MPC 02	1561	SD	22.4	1755	SD	52.0	
MPC 03	1517	%CV	1.5	1856	%CV	2.9	
LPC 01	207	Mean	208	278.5	Mean	273	-23.8
LPC 02	207	SD	1.73	264.0	SD	7.97	
LPC 03	210	%CV	0.8	277.0	%CV	2.9	

PC: LT3114 anti-idiotypic rabbit antibody dilutions in pooled Rat Serum (HPC 1:100, LPC 1:400 & LPC 1:3200)

Run ID: 140711-3441402-RTSE-RLM-3C-009

%Difference: $100 * ((\text{Mean 24 hour treated PC} - \text{Mean untreated PC}) / (\text{Mean untreated PC}))$

RU: Response Unit

Table 9B Room Temperature (24 hour) Stability of anti-LT3114 anti-idiotypic rabbit antibodies in Rat Serum (Repeat)

PC Level	Treated (24hr RT) / RU values			Untreated /RU Values			% Difference
HPC 01	8435	Mean	8772	9358	Mean	7405	18.5
HPC 02	8827			9426			
HPC 03	8367			6188			
HPC 04	8505	SD	579	6373	SD	1545	-22.9
HPC 05	8592	%CV	6.6	6578	%CV	20.9	
HPC 06	9908			6507			
LPC 01	201.5	Mean	219	290.0	Mean	284	-22.9
LPC 02	222.0			284.5			
LPC 03	227.0			282.0			
LPC 04	219.5	SD	9.11	298.0	SD	10.4	-22.9
LPC 05	221.5	%CV	4.2	285.5	%CV	3.7	
LPC 06	224.5			266.5			

PC: LT3114 anti-idiotypic rabbit antibody dilutions in pooled Rat Serum (HPC 1:100, LPC 1:400 & LPC 1:3200)

Run ID: 140716-3441402-RTSE-RLM-1A-010

%Difference: $100 * ((\text{Mean 24 hour treated PC} - \text{Mean untreated PC}) / (\text{Mean untreated PC}))$

RU: Response Unit

Table 9C Room Temperature (7 hour) Stability of anti-LT3114 anti-idiotypic rabbit antibodies in Rat Serum

PC Level	Treated (7hr RT) / RU values			Untreated /RU Values			% Difference
HPC 01	8580	Mean	8915	9358	Mean	8324	7.1
HPC 02	8911	SD	337	9426	SD	1850	
HPC 03	9253	%CV	3.8	6188	%CV	22.2	
MPC 01	1755	Mean	1744	1942	Mean	1906	-8.5
MPC 02	1709	SD	31.4	1899	SD	33.5	
MPC 03	1769	%CV	1.8	1876	%CV	1.8	
LPC 01	222.5	Mean	236.7	290.0	Mean	285.5	-17.1
LPC 02	240.0	SD	12.8	284.5	SD	4.09	
LPC 03	247.5	%CV	5.4	282.0	%CV	1.4	

PC: LT3114 anti-idiotypic rabbit antibody dilutions in pooled Rat Serum (HPC 1:100, LPC 1:400 & LPC 1:3200)

Run ID: 140716-3441402-RTSE-RLM-1A-010

%Difference: $100 * ((\text{Mean 7 hour treated PC} - \text{Mean untreated PC}) / (\text{Mean untreated PC}))$

RU: Response Unit

16. APPENDICES

Appendix A: Datasheet of LT3114 Drug

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416
Formulation Buffer		Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)		2 to 8 °C
Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

Summary
<u>Production location, date, and method</u> The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C. Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage. Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage. Purified lots LP121416 and LP121417 were pooled to create lot LP121419. <u>Purification location, date and method</u> Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning). <u>Concentration by A₂₈₀ and extinction coefficient</u> The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg. <u>Total amount of Ab (mg)</u> 1400 mg <u>Appearance details:</u> Clear, slightly yellow solution

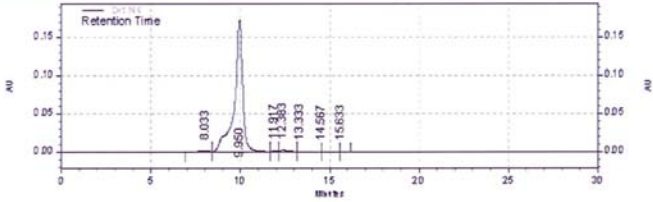
Confidential

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	Oct 31, 2013	Nov 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Two pronounced bands; One at ~55 kDa; One at ~ 28 kDa
	<i>Insert images of SDS-PAGE</i>	Non-Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty 	Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty 
	Isoelectric Focusing (by IEF Gel)	Report Results	Not done

Confidential

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	3 of 5

Attribute	Test	Specifications	Results																																													
Potency	Protein Concentration (OD-280)	Report Results	20.14 mg/mL																																													
	LPA Binding Direct ELISA (EC 50)	Report Results	25.05 ng/mL																																													
	Bioactivity Assay (IL-6 and EDG2)	Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 μg/mL																																													
Purity	Insert image of SE-HPLC <table><caption>Det 166 Results</caption><thead><tr><th>Time</th><th>Area</th><th>Area %</th><th>Height</th><th>Height %</th></tr></thead><tbody><tr><td>8.033</td><td>68097</td><td>1.14</td><td>1413</td><td>0.78</td></tr><tr><td>9.950</td><td>5707101</td><td>95.95</td><td>173377</td><td>96.24</td></tr><tr><td>11.917</td><td>35431</td><td>0.60</td><td>1744</td><td>0.97</td></tr><tr><td>12.383</td><td>97022</td><td>1.63</td><td>2645</td><td>1.47</td></tr><tr><td>13.333</td><td>31081</td><td>0.52</td><td>668</td><td>0.37</td></tr><tr><td>14.567</td><td>8116</td><td>0.14</td><td>234</td><td>0.13</td></tr><tr><td>15.633</td><td>1015</td><td>0.02</td><td>62</td><td>0.03</td></tr><tr><td>Totals</td><td>5947863</td><td>100.00</td><td>180143</td><td>100.00</td></tr></tbody></table>			Time	Area	Area %	Height	Height %	8.033	68097	1.14	1413	0.78	9.950	5707101	95.95	173377	96.24	11.917	35431	0.60	1744	0.97	12.383	97022	1.63	2645	1.47	13.333	31081	0.52	668	0.37	14.567	8116	0.14	234	0.13	15.633	1015	0.02	62	0.03	Totals	5947863	100.00	180143	100.00
	Time	Area	Area %	Height	Height %																																											
	8.033	68097	1.14	1413	0.78																																											
9.950	5707101	95.95	173377	96.24																																												
11.917	35431	0.60	1744	0.97																																												
12.383	97022	1.63	2645	1.47																																												
13.333	31081	0.52	668	0.37																																												
14.567	8116	0.14	234	0.13																																												
15.633	1015	0.02	62	0.03																																												
Totals	5947863	100.00	180143	100.00																																												
SE-HPLC	Purity [as % IgG monomer]	≥ 90%	96%																																													
	Aggregates [as % of total peak area]	≤ 6%	1%																																													
	Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%																																													

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		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	Oct 31, 2013	Nov 1, 2013	4 of 5

Attribute	Test		Specifications	Results
	SDS-PAGE reduced	Purity [%]	≥ 90%	≥ 90%
		50 kDa Band	~65%	~67%
		25 kDa Band	~35%	~33%
		Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin		≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden		≤ 10 CFU/mL	Not done

Comments																																
Source Data: <table border="1"> <tr> <th></th> <th>Lpath notebook</th> <th>Pages</th> </tr> <tr> <td>Production</td> <td>277</td> <td>149</td> </tr> <tr> <td>Purification</td> <td>296</td> <td>11,18,25</td> </tr> <tr> <td>UV A280</td> <td>296</td> <td>48</td> </tr> <tr> <td>SDS-PAGE</td> <td>296</td> <td>26</td> </tr> <tr> <td>LPA binding ELISA</td> <td>221</td> <td>21-22</td> </tr> <tr> <td>IL-6</td> <td>237</td> <td>48-49</td> </tr> <tr> <td>EDG2</td> <td>237</td> <td>51</td> </tr> <tr> <td>SE-HPLC</td> <td>296</td> <td>33-34</td> </tr> <tr> <td>Endotoxin</td> <td>298</td> <td>13</td> </tr> </table>				Lpath notebook	Pages	Production	277	149	Purification	296	11,18,25	UV A280	296	48	SDS-PAGE	296	26	LPA binding ELISA	221	21-22	IL-6	237	48-49	EDG2	237	51	SE-HPLC	296	33-34	Endotoxin	298	13
	Lpath notebook	Pages																														
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IL-6	237	48-49																														
EDG2	237	51																														
SE-HPLC	296	33-34																														
Endotoxin	298	13																														
Additional Comments on Sample Quality:																																

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Appendix B: Datasheet of Anti-LT3114 Whole Rabbit Serum



May 20, 2014

Rupalie Meegalla
QPS, LLC
1 Innovation Way, Suite 200
Newark, DE19711

Dear Rupalie,

Enclosed, please find the following materials in this shipment:

Whole rabbit serum from bleed 2 of rabbit immunized with whole 3114 IgG
aliquoted in 2 vials 1ml/each (Whole 3114 anti-idiotypic)

Upon arrival, please stored at 4° C. Material stored at 4° C under sterile condition is
stable for up to 2 months.

Should you have any questions regarding the material, please do not hesitate to contact
us.

Best regards,

A handwritten signature in black ink, appearing to read "Rosalia Matteo".

Rosalia Matteo, PhD
Asst. Director
Lpath Incorporated

4025 Sorrento Valley Blvd., San Diego, CA 92121-1404 • Main: (858) 678-0800 • Fax: (858) 678-0900
www.Lpath.com

Appendix C Statistical Analysis for Screening Assay Cut Point Factor and Confirmatory Cut Point Outlier Evaluation

All statistical analyses were completed using JMP Statistical Discovery Software (Version 10.0; SAS Institute, Inc., Cary, NC, USA). Statistical methods used for the analysis are consistent with procedures recommended by Shankar G, et al. 2008.

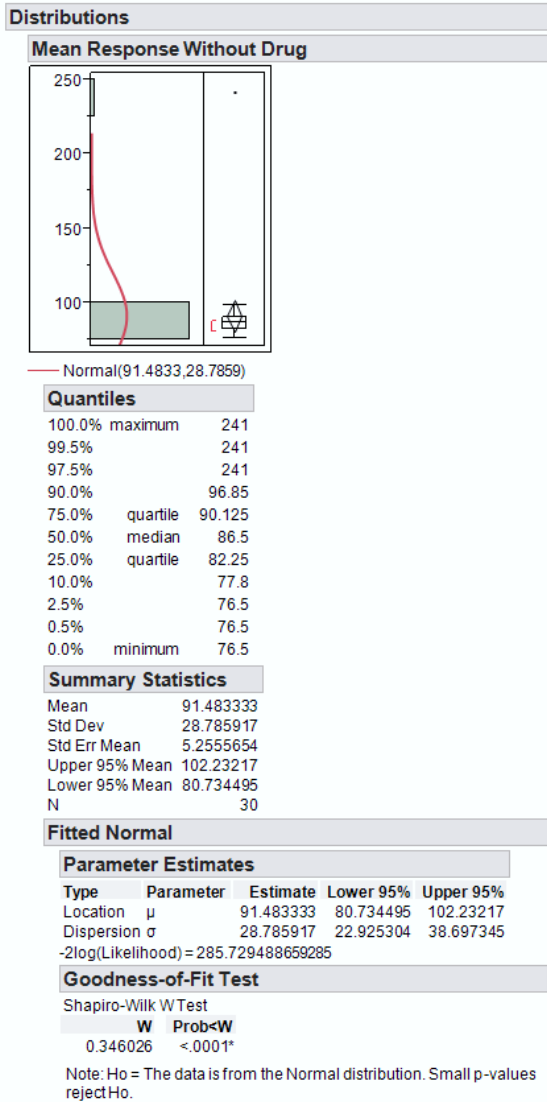
The average RU values of individual samples were imported into JMP. Readings that had %CV greater than 25% were excluded.

First, statistical outliers for screening analysis were evaluated on a per-day basis using the “box-plot” method in JMP software. The detailed formula is also listed in QPS SOP, QPS-TLM-001. Individuals that have a response above the upper boundary or below the lower boundary were exported as an Excel file and excluded from further analysis. Excel file is included for verification of outliers ([Table 4a](#)).

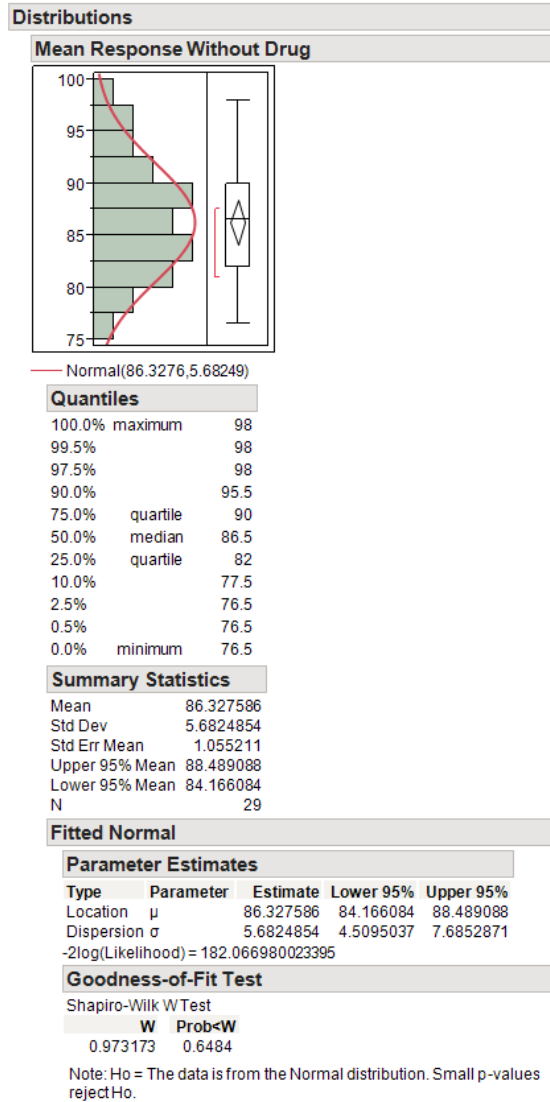
Next, the normality for screening analysis was assessed on a per-day basis using Shapiro-Wilk test in JMP software. The normality was indicated by the probability in the “Goodness-of-Fit”. If the probability is greater or equal to 0.05, the data is normally distributed. If the probability is less than 0.05, the data is not normally distributed.

Additionally, statistical outliers for confirmatory analysis were evaluated using the “box-plot” method in JMP software ([Table 4b](#)).

Day 1 (Initial)

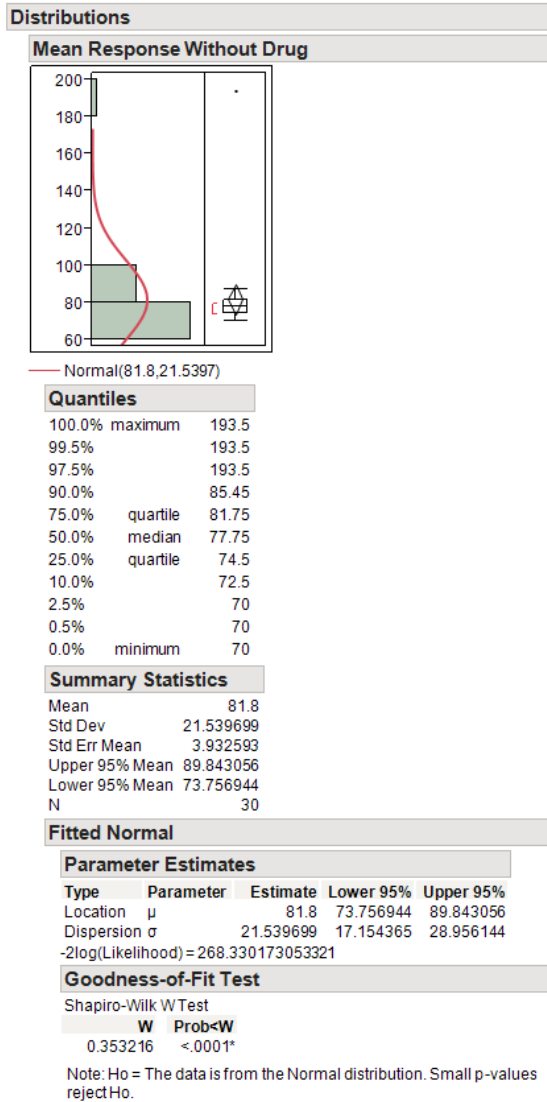


Day 1 (Final)

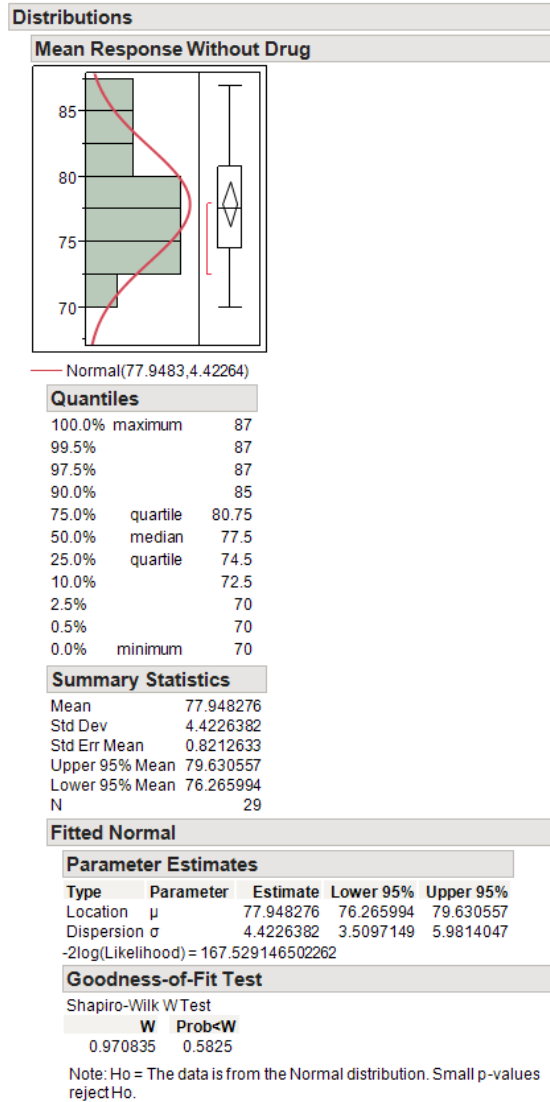


Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.6484).

Day 2 (Initial)



Day 2 (Final)

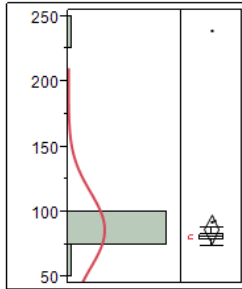


Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.5825).

Day 3 (Initial)

Distributions

Mean Response Without Drug



Normal(86.4667,29.1104)

Quantiles

100.0%	maximum	239
99.5%		239
97.5%		239
90.0%		91.6
75.0%	quartile	82.625
50.0%	median	80.5
25.0%	quartile	78.875
10.0%		76.55
2.5%		74
0.5%		74
0.0%	minimum	74

Summary Statistics

Mean	86.466667
Std Dev	29.110353
Std Err Mean	5.314799
Upper 95% Mean	97.336651
Lower 95% Mean	75.596682
N	30

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	86.466667	75.596682	97.336651
Dispersion	σ	29.110353	23.183687	39.133489
-2log(Likelihood) = 286.40194475136				

Goodness-of-Fit Test

Shapiro-Wilk W Test

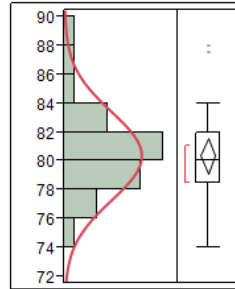
W	Prob<W
0.290210	<.0001*

Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Day 3 (Final)

Distributions

Mean Response Without Drug



Normal(80.3889,3.06448)

Quantiles

100.0%	maximum	88
99.5%		88
97.5%		88
90.0%		84.7
75.0%	quartile	82
50.0%	median	80
25.0%	quartile	78.5
10.0%		76.4
2.5%		74
0.5%		74
0.0%	minimum	74

Summary Statistics

Mean	80.388889
Std Dev	3.064478
Std Err Mean	0.5897591
Upper 95% Mean	81.601156
Lower 95% Mean	79.176622
N	27

Fitted Normal

Parameter Estimates

Type	Parameter	Estimate	Lower 95%	Upper 95%
Location	μ	80.388889	79.176622	81.601156
Dispersion	σ	3.064478	2.4133268	4.1996576
-2log(Likelihood) = 136.096052676496				

Goodness-of-Fit Test

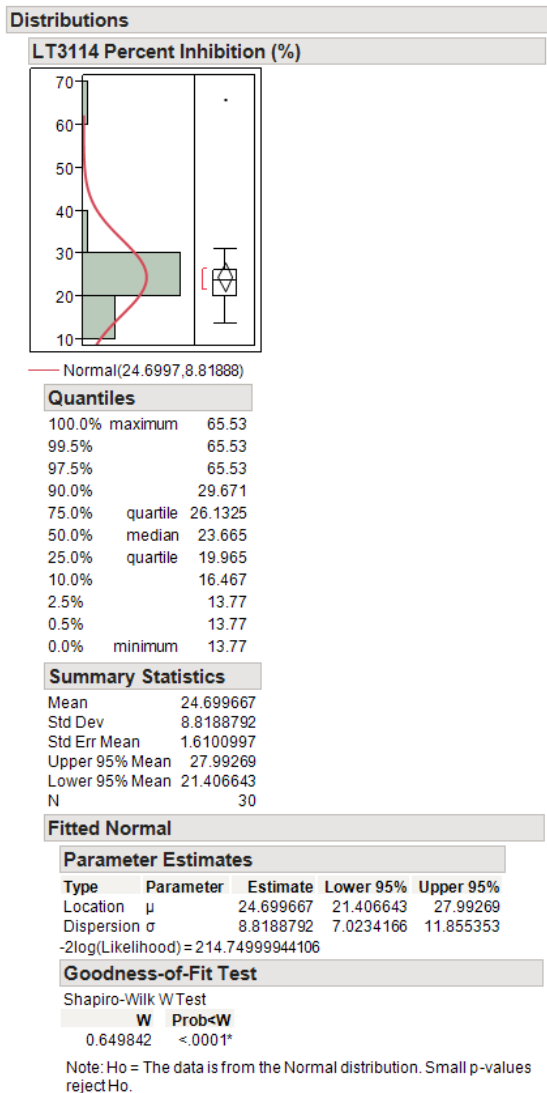
Shapiro-Wilk W Test

W	Prob<W
0.950532	0.2209

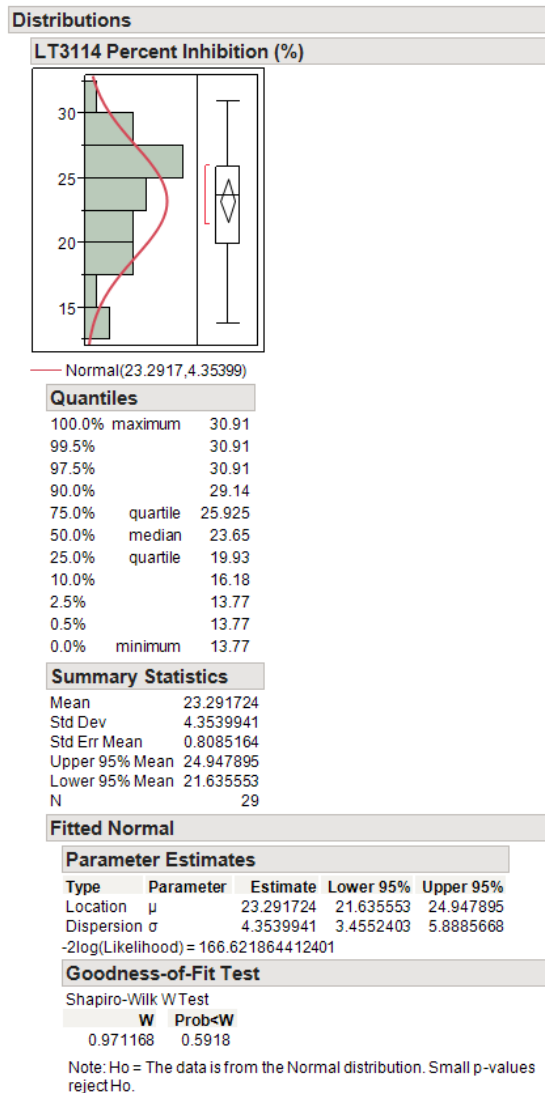
Note: Ho = The data is from the Normal distribution. Small p-values reject Ho.

Data distribution was assessed using JMP Statistical Discovery with the assumption that the data is normally distributed. The model of assumption was confirmed by Shapiro-Wilk W Test (P value = 0.2209).

Confirmatory Assay (Initial)



Confirmatory Assay (Final)



One outlier was identified based on the box-plot approach using JMP Statistical Discovery for confirmatory assay cut point determination.

Appendix D Analytical Procedure for the Detection of Anti- LT3114 Antibodies in Rat Serum by ECL

QPS, LLC

ANALYTICAL PROCEDURE

TITLE: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

QPS Project No.: _____ **Run ID:** _____
Client Study No.: _____

Page: 1 of 10

The analytical procedure is approved for sample analysis? ☒ Yes ☐ No

If Yes, then all validation data tables (i.e. except SSS & LTFS) must be compiled & signed off by PI

Procedure Reviewed and Approved by

R. Meegalla
Principal Investigator (PI)

Date: 7/24/14

Procedure Reviewed and Approved by

[Signature]
Scientific Review

Date: 7/24/14

Procedure Reviewed by

[Signature]
Quality Assurance
344-1402-01
Analytical Procedure Tracking Number

Date: 7-24-14

Copy made for QPS Project Number: _____

***Distribution Initials/Date:** _____ **Copy Number:** _____

**Analytical procedure expires 7 days from the distribution date*

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Page 1 of 10
344-1402-01

Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

1. Objective:

Sample analysis: ☐ Sample screening ☐ Positive sample confirmation

Validation: (List Test)

2. Materials and Equipment (as specified or equivalent):

MSD Standard Coated Streptavidin plate: Cat# L11SA-2 or equivalent

Plate Sealer: Costar Cat.# 3095

25-100 mL Reagent Reservoir: Costar Cat.# 4870

96-well format micro titer/dilution tubes (VWR Cat.# 46300-536)

Polypropylene dilution plate (Nunc Cat.# 267385)

0.5-50 mL polypropylene tubes, VWR Scientific

500 or 1000 mL wash bottle

Graduated Cylinder

Multi-tube vortexer: VWR Scientific Products. Equipment # _____

Balance: Equipment # _____

Plate Shaker: Equipment # _____

Microplate reader: Meso Scale Discovery Sector Imager 6000: Equipment # _____

Pipette Identification: Provide ID numbers for each listed:

20 μ L Pipette ID# _____

100 μ L Pipette ID# _____

200 μ L Pipette ID# _____

1000 μ L Pipette ID# _____

Multichannel Pipette ID# _____

Initials

Date

3. Specific Assay Materials and Reagent Information:

<u>Reagents or Chemical/Concentration</u>	<u>Lot#</u>	<u>Exp. Date</u>	<u>NB#</u>
Biotinylated LT3114 (633 μ g/mL)	<u>RLM140522-02</u>	<u>4/30/16</u>	<u>LT3114 Assay Notes</u>
Ruthenylated LT3114 (753 μ g/mL)	<u>RLM140522-03</u>	<u>4/30/16</u>	<u>LT3114 Assay Notes</u>

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

<u>Reagent or Chemical (Cont.)</u>	<u>Lot#</u>	<u>Exp. Date</u>	<u>NB#</u>
LT3114(HU-anti-LPA antibody) (19.33 mg/mL)	LP121416	4/30/16	Client 344 Compound Receipt NB
anti-LT3114 positive Rabbit Serum	R-2265-B2-11/13	5/21/16	Client 344 Reagent Receipt NB
Streptavidin coated MSD Plate (MSD Cat# L11SA-2)	_____	_____	NA
BSA Fraction V(Fatty Acid Free) (Calbiochem Cat #126575 or equivalent)	_____	_____	NA
4X MSD Read Buffer T (MSD Cat# R92TC-1)	_____	_____	NA
Water, Type I, from Elgastat UHQ-PS, Elga Ltd. or equivalent			

Verified and Entered by: _____

Initials Date

4. Buffer Preparation

	<u>Lot#</u>	<u>Exp. Date</u>	<u>NB#</u>
a) PBST Buffer: (0.05% Tween-20 in 1X PBS) (500 µL of Tween 20 into 1 L of 1x PBS)	_____	_____	_____

b) **Assay Buffer:** (1% BSA in PBST) prepare fresh and discard after use.

Circle applicable preparation:

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
BSA	0.20 g	0.32 g	0.48 g	0.64 g	
PBST	20 mL	32 mL	48 mL	64 mL	
Total Volume	20 mL	32 mL	48 mL	64 mL	

c) **Blocking Buffer:** (4% BSA in 1X PBST Buffer)

Circle applicable preparation:

	1 Plate	2 Plates	3 Plates	4 Plates	See attached worksheet for alternative preparation
BSA	0.64 g	1.28 g	1.92 g	2.56 g	
PBST	16 mL	32 mL	48 mL	28 mL	
Total Volume	16 mL	32 mL	48 mL	64 mL	

Initials Date

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

5. Positive/Negative Controls

Positive Controls (PC): Anti-LT3114 positive Rabbit Serum diluted in RTSE

PC Level	N=	Lot #	Exp. Date	NB
HPC (1:100 in RTSE)				Client 344 LT3114 Assay Notes
MPC (1:400 in RTSE)				
LPC (1:3200 in RTSE)				

Storage temperature °C	Time removed from storage location	
	Time returned to storage location or <u>check</u> discarded if not returned	☐ Discarded

Negative Control (NC): Pooled Rat serum (n= __, duplicate measurement)

Lot #: _____

Exp. Date: _____

NB#: _____

Initials

Date

6. Validation Information

6.1. Determination of Screening Cut Point Factor (n= __)

☐ See attached sheet for information of individual Rat Serum lots

6.2. Determination of Confirmatory Assay Cut Point Factor (n= __)

☐ See attached sheet for information of individual Rat Serum lots

6.3. Drug Tolerance Test (n= __)

☐ See attached worksheet for PC and Drug LT3114 Dilutions.

6.4. Assay Selectivity (Recovery) Test

☐ See attached worksheet for information on individual RTSE lots

6.5. Room Temperature Stability of Positive Control (n= __)

PC Level	Lot #	Exp. Date	NB
HPC (1:100 in RTSE)			Client 344 LT3114 Assay Notes
MPC (1:400 in RTSE)			
LPC (1:3200 in RTSE)			

6.6. Freeze/Thaw Stability (n= __) Stored at -70°C # of Cycles _____

PC Level	Lot #	Exp. Date	NB
HPC (1:100 in RTSE)			Client 344 LT3114 Assay Notes
MPC (1:400 in RTSE)			
LPC (1:3200 in RTSE)			

☐ Refer to QPS Stability Forms for Further Information

Initials

Date

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

7. Sample Information (see Plate Map)

Note: See Plate Map and associated Sample Custody Tracking Tag (QPS Form 177)

Length of established Room Temperature Stability: 24 hours

Total number of samples to be analyzed: _____

Initials

Date

Re-analysis Sample Summary

☐ See attached sheet for a list of re-assay samples

- If whole plate is being re-analyzed, give current, previous Run IDs and the reason.

Current Run ID:

Previous Run ID:

Reason for Re-analysis:

- Specify sample IDs and reason for re-assay

Client instruction _____

Other _____

Initials

Date

Management authorization for client requested re-assay:

Initials

Date

Sample Sequence/ Position verified by:

(Verification must be done by another staff member and prior to addition of samples to plate)

Initials

Date

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

8.0 ECL ASSAY PROCEDURE:

Note: Analyst must have been trained for ECL assays and hand/machine washing technique.

8.1. Block Plate/Plates:

Block Streptavidin coated MSD plate with 150 μ L/well of Blocking Buffer and incubate for at least 60 minutes at room temperature on a plate shaker (setting 3-4).

Plate:	<u>1A</u>	<u>2B</u>	<u>3C</u>	<u>4D</u>
Start:	_____	_____	_____	_____
End:	_____	_____	_____	_____
MSD Barcode:	_____	_____	_____	_____

Initials

Date

8.2. Sample Preparation for Screening:

a) Arrange dilution tubes according to attached plate map. Use one tube for duplicate measurement of one sample.

b) Prepare a 1:30 dilution of each study sample in Assay buffer:

10 μ L study sample + 290 μ L Assay Buffer

8.3. Sample Preparation for Confirmation:

a) Arrange dilution tubes according to attached plate map. Use one tube for duplicate measurement of one sample.

b) Prepare LT3114 spiked assay buffer at 100ug/mL concentration.

Circle applicable preparation:

	1 Plate 40 μ L	2 Plates 80 μ L	3 Plates 100 μ L	4 Plates 125 μ L	See attached worksheet for alterna- tive prepa- ration
LT3114 stock A (19.33 mg/mL)					
Assay Buffer	7.69 mL	15.38 mL	19.23 mL	24.04 mL	
Total Volume	7.73 mL	15.46 mL	19.33 mL	24.16 mL	

c) Prepare 1:30 dilutions of each study sample / HPC/MPC/ LPC in Drug Spiked Assay buffer

10 μ L study sample/NC/PC + 290 μ L LT3114 Spiked Assay Buffer

d) Incubate diluted confirmation samples for at least 30 minutes at room temperature.

Plate:	<u>1A</u>	<u>2B</u>	<u>3C</u>	<u>4D</u>
Start:	_____	_____	_____	_____
End:	_____	_____	_____	_____

Initials

Date

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8.4. Master Mix Preparation:

- a) Prepare Biotinylated LT3114 Stock B (20 µg/mL) in Assay Buffer:
20 µL of Biotinylated LT3114 Stock (633 µg/mL) + 613 µL Assay Buffer
- b) Prepare Ruthenylated LT3114 Stock B (20 µg/mL) in Assay Buffer:
20 µL of Ruthenylated LT3114 Stock (753 µg/mL) + 733 µL Assay Buffer
- c) Prepare 500 ng/mL Biotinylated and Ruthenylated LT3114 in Assay Buffer:

Circle applicable preparation:

	<u>1 Plate</u>	<u>2 Plates</u>	<u>3 Plates</u>	<u>4 Plates</u>
Biotinylated LT3114 (20 µg/mL)	150 µL	300 µL	400 µL	550 µL
Ruthenylated LT3114 (20 µg/mL)	150 µL	300 µL	400 µL	550 µL
Assay Buffer	5.7 mL	11.4 mL	15.2 mL	20.9 mL
Total Volume	6 mL	12 mL	16 mL	22 mL

Initials

Date

8.5. Sample Loading to Polypropylene Plate:

- a) Add 50 µL of each diluted PC, NC and sample to the polypropylene plate according to the attached plate map (one well per duplicate measurement).
- b) Add 100 µL/well of the Master Mix Solution to the polypropylene dilution plate (one well per duplicate measurement).
- c) Cover plate with a plate sealer and incubate 1 hour ±10 minutes in the **dark** at room temperature on a plate shaker (setting 3-4).

	<u>1A#</u>	<u>2B#</u>	<u>3C#</u>	<u>4D#</u>
Start:	_____	_____	_____	_____
End:	_____	_____	_____	_____

Initials

Date

8.6. Sample Mixture Transfer:

- a) Wash Streptavidin coated MSD plate **1X** with PBST wash buffer.
- b) Transfer 50 µL of the incubated sample mixture from the polypropylene plate into duplicate wells of the MSD plate (50µL/well).
- c) Cover plate and incubate the plate for 1 hour (±10 minutes) **in the dark** on a plate shaker (setting 3-4).

Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

1A# **2B#** **3C#** **4D#**

Start: _____

End: _____

Initials Date

8.7. Prepare and Add 2X Read Buffer T:

- a) Prepare 2X Read Buffer T by mixing equal volumes of 4X Read Buffer T with Type I water.

Circle applicable preparation:

	1 Plate	2 Plates	3 Plates	4 Plates
4X Read Buffer	8.5 mL	16 mL	23 mL	30 mL
Type I Water	8.5 mL	16 mL	23 mL	30 mL
Total Volume	17 mL	32 mL	46 mL	60 mL

- b) Wash MSD plate **3X** with PBST

- c) Add **150µL/well** 2X Read Buffer T to the MSD plate

- d) Read plate on MSD Sector Imager 6000. Name the file after the run ID: "RunID.txt"

Initials Date

8.8. Data Processing:

- a) Print MSD Screenshot and note with Run IDs as raw data

- b) Check the following if applicable:

☐ Discovery Workbench™ 2006 MSD_3_0_17_3

☐ Data transferred to Watson

Initials Date

9. PROCEDURE REVIEW

Analyst(s) performing assay:

Initials Date

Data processed/finalized by:

Initials Date

Peer Review

Initials Date

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

10. PRINCIPAL INVESTIGATOR APPROVAL

- 10.1. Determine pass/fail of each analytical run according to the acceptance criteria specified in QPS 344-1402 validation report and QPS-TLM-001.

Run	Pass (P) / Fail (F) (if F, complete section 10.3)

10.2. Validation

If applicable, do test(s) during the validation meet the acceptance criteria as specified in the SOPs (e.g. Stability Test(s), Specificity, Other)?

Yes / No / N/A (Circle one) (if no, complete section 3)

If applicable, should Study Samples be re- assayed? Decision is determined from the data interpreted today.

Yes / No / N/A (Circle one)

List sample identification and reason for re-assay (☐ see Watson All Samples Table)

Principal Investigator

Initials

Date

10.3. Review the following to:

☐ Determine root cause of failed run or test (if applicable)

☐ Investigate experimental observations: _____

☒ = Reviewed & evaluated ☐ = Not applicable (NA)

- ☐ Analytical procedure documentation:
- ☐ Positive control documentation:
- ☐ Negative control documentation:
- ☐ Solution preparation (buffer, etc.):
- ☐ Matrices:
- ☐ Plate preparation (lot#, Ab, #days, etc.):
- ☐ Washing method:
- ☐ Analyst familiarity with method:

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Title: Analytical Procedure for the Detection of Anti-LT3114 antibodies in Rat Serum by ECL (V01)

- ☐ Plate read setup:
- ☐ Master Mix preparation and incubation:
- ☐ Read buffer T preparation:
- ☐ Employee training records:
- ☐ Analyst laboratory error:
- ☐ Equipment maintenance log:

Conclusion: _____

Principal Investigator

Initials

Date

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